

FUNCTIONAL REQUIREMENTS SPECIFICATION

PSR — Policy & System Reform Platform
Ethiopian Ministry of Health

Document Title	Functional Requirements Specification — PSR Platform
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This document defines the functional and non-functional requirements for the PSR digital platform — a centralized system for the Ethiopian Ministry of Health and its agencies to manage health policy documents and research proposals end-to-end.

1. Project Scope

The PSR platform will serve as the authoritative digital backbone for health policy and research governance within the Ethiopian Ministry of Health ecosystem. The following scope boundaries define what the system will — and will not — deliver in version 1.0.

1.1 In-Scope

Policy Document Lifecycle Management

- Submission, review, and approval workflows for policy concept notes
- Multi-stage draft development pipeline with versioning and expert checklist scoring
- Central policy repository with serial number generation, version control, and effective/review date tracking
- Formal revision processes for post-implementation changes, preserving historical 'outdated' versions
- Revision feedback loops supporting continuous policy improvement

Research Management Workflow

- Management of Call for Proposals — budget years, deadlines, poster attachments
- Structured proposal submission for Principal Investigators (PIs) and Co-PIs including executive summaries and thematic alignment
- PSR Office initial screening and Research Advisory Committee (ROC) technical evaluation using scoring checklists
- Monitoring & progress tracking — periodic 6-month reports, ethical clearance, and IRB application tracking
- Final output submission — mandatory policy briefs and full research reports with NDMC-EPHI data deposit integration

Core System & Administrative Tools

- Role-Based Access Control (RBAC) with granular permissions per user type
- User profile management — education levels, academic ranks, and organizational affiliations
- OTP/2FA authentication, trusted IP tracking, and session management
- Centralized taxonomy management — organizations, units, thematic areas, strategic objectives
- Full-text search across the repository by keywords, document types, years, and thematic areas
- Comprehensive audit logging — user actions, module changes, timestamps
- Dashboard with summary KPIs, workflow status views, and quick navigation
- Notification system — in-app and configurable alerts for workflow events

1.2 Out-of-Scope (Version 1.0)

#	Out-of-Scope Item	Rationale / Future Consideration
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1	Financial & budget management for research grants	Requires integration with MoH financial systems — Phase 2
2	External public portal for citizens to view policy documents	Separate public-facing product decision required
3	Real-time video conferencing or meeting integration	Third-party service integration deferred to Phase 2
4	Mobile native applications (iOS / Android)	Web-responsive first; native apps are Phase 3
5	AI-assisted document summarization or analysis	Requires model procurement and data governance approval
6	Integration with external research databases (PubMed, WHO, etc.)	API availability and data agreements pending
7	Automated translation between Amharic and English	Language processing requires separate NLP stack
8	eSignature or legally binding digital approvals	PKI/legal framework not yet in place at MoH

2. User Personas

Six primary personas have been identified based on the stakeholder analysis. Each persona description covers role, primary goals, key pain points, and system access level.

P-01 Dr. Aisha — PSR/LEO Office Official	ROLE: Ministry of Health PSR Office senior official overseeing policy and research governance. Goals - Maintain visibility over all active policy and research workflows - Approve or reject concept notes, drafts, and proposals - Generate performance reports on policy development progress Pain Points - Currently relies on email chains and manual tracking spreadsheets - No audit trail for who approved what and when - Cannot easily see bottlenecks in multi-stage approval pipelines System Access: Full RBAC access to Policy Lifecycle, Research Management, Reports, and Admin modules.
P-02 Researcher Bekele — Policy/Research Developer	ROLE: University-affiliated researcher submitting research proposals and policy concept notes. Goals - Submit proposals quickly with clear status feedback - Track review progress and reviewer comments in one place - Upload progress reports and final outputs without manual follow-up Pain Points - Uncertainty about submission status for weeks after submitting - Different format requirements communicated informally - No structured way to respond to reviewer feedback System Access: Can create and manage own submissions; read-only on others; receives automated notifications.
P-03 Expert Tigist — Technical Reviewer (ROC Member)	ROLE: Member of the Research Advisory Committee evaluating proposals and policy drafts. Goals - Access assigned documents for review with all relevant context - Complete structured scoring checklists efficiently - Provide comments that feed back to the submitter Pain Points - Documents currently shared via email, often missing supporting files - No standard scoring template, leading to inconsistent evaluations

- Difficulty coordinating comments with other reviewers

System Access: Access only to documents assigned for review; can score and comment; cannot modify submissions.

P-04

Admin Selam — System Administrator

ROLE: IT/operations personnel managing the PSR platform configuration and user accounts.

Goals

- Onboard and manage user accounts, roles, and permissions
- Manage taxonomy — organizations, units, thematic areas
- Monitor audit logs and system health

Pain Points

- No centralized user management — handled across multiple systems
- Cannot easily audit who accessed or changed what
- Role changes require direct database intervention

System Access: Full system administration access; no submission or review capability by default.

P-05

Director Yonas — Senior Leadership / Decision Maker

ROLE: Senior MoH director consuming dashboard reports and repository analytics.

Goals

- High-level visibility into policy and research pipeline status
- Access the approved policy repository quickly
- Track compliance with review timelines

Pain Points

- Reporting is manually compiled and delayed
- No single source of truth for approved policies
- Cannot drill into specific workflow bottlenecks without staff support

System Access: Read-only dashboard, repository, and reporting access.

P-06

Partner Rep. Mekdes — Institutional / NGO Partner

ROLE: Representative from a partner university or NGO participating in research calls.

Goals

- Respond to Call for Proposals and submit required documents
- Track the status of submitted proposals
- Receive structured feedback after review

Pain Points

- Call for Proposals are communicated informally via email
- Proposal submission requirements are unclear



- No transparency on timeline after submission

System Access: Limited to own submissions and public Call for Proposals module.

3. Functional Requirements (MoSCoW)

MUST HAVE	Critical requirements that define the core product. The system cannot go live without these.
SHOULD HAVE	High-value requirements that significantly improve the platform but are not launch-blockers.
COULD HAVE	Enhancement features that add value if capacity allows; deferred to a later release if necessary.

3.1 Authentication & User Management

ID	Category	Requirement	Priority
FR-1.1	Authentication	Users must authenticate via phone number and password at a minimum.	MUST
FR-1.2	Authentication	OTP/2FA must be supported for login and sensitive actions (password reset, admin operations).	MUST
FR-1.3	Session	JWT access and refresh token lifecycle must be implemented with automatic silent refresh.	MUST
FR-1.4	Session	Sessions must be invalidated on explicit logout; refresh tokens must be blacklisted.	MUST
FR-1.5	User Profile	Each user account must include: name, phone, email, sex, title, organization, unit, and status.	MUST
FR-1.6	User Profile	Educational level, academic rank, and research area must be captured for researcher profiles.	MUST
FR-1.7	RBAC	The system must enforce role-based access control; permissions must be grouped by role and resource.	MUST
FR-1.8	RBAC	Roles must support granular permissions: view, create, edit, delete, approve per module.	MUST
FR-1.9	Security	Trusted IP addresses must be trackable per user for anomaly detection.	SHOULD
FR-1.10	Audit	All login attempts (success and failure), role changes, and account status changes must be logged.	MUST
FR-1.11	User Mgmt	Admins must be able to activate, deactivate, and reset passwords for any user account.	MUST
FR-1.12	Notification	Users should receive in-app and configurable email/SMS notifications for account events.	SHOULD
FR-1.13	Profile	Users could upload a profile photo; file type and size must be validated.	COULD

3.2 Policy Document Lifecycle Management

ID	Category	Requirement	Priority
FR-2.1	Concept Note	Authorized users must be able to submit a new policy concept note with title, thematic area, rationale, and file attachments.	MUST
FR-2.2	Concept Note	Concept notes must be automatically versioned; prior versions must be preserved and auditable.	MUST
FR-2.3	Concept Note	PSR officials must be able to review, approve, request revision, or reject concept notes.	MUST
FR-2.4	Concept Note	Submitters must receive status notifications at each stage transition.	MUST
FR-2.5	Draft Review	Approved concept notes must transition to draft development with multi-reviewer assignment.	MUST
FR-2.6	Draft Review	Reviewers must complete structured scoring checklists during draft evaluation.	MUST
FR-2.7	Draft Review	Aggregate review scores and individual comments must be visible to the PSR office.	MUST
FR-2.8	Draft Review	Draft revisions must support multiple rounds with tracked changes between versions.	SHOULD
FR-2.9	Repository	Approved policy documents must be registered in the central repository with a system-generated serial number.	MUST
FR-2.10	Repository	Repository entries must record: effective date, review date, document type, version, and owner.	MUST
FR-2.11	Repository	The system must support revision of published documents; outdated versions must be archived, not deleted.	MUST
FR-2.12	Repository	Repository search must support filters: keyword, document type, thematic area, year, organization.	MUST
FR-2.13	Reporting	A policy pipeline summary report (concept notes → drafts → approved → published) must be exportable as PDF/Excel.	SHOULD
FR-2.14	Workflow	Document checklist items must be configurable by document type in the admin settings module.	SHOULD
FR-2.15	Collaboration	Policy stakeholders could annotate/comment on draft documents inline before formal submission.	COULD

3.3 Research Management Workflow

ID	Category	Requirement	Priority
FR-3.1	Call for Proposals	PSR administrators must be able to create and publish research Calls for Proposals with budget year, deadline, scope, and attachments.	MUST

FR-3.2	Call for Proposals	Calls must be visible to eligible researchers based on their organizational and thematic area profile.	MUST
FR-3.3	Proposal Submission	Researchers must be able to submit proposals linked to an open Call, including: title, PI/Co-PI details, executive summary, thematic area, and methodology.	MUST
FR-3.4	Proposal Submission	Proposals must support file attachments (proposal document, budget, ethics clearance).	MUST
FR-3.5	Screening	The PSR Office must be able to perform initial administrative screening before forwarding to ROC.	MUST
FR-3.6	ROC Review	ROC members must be assigned to specific proposals and complete structured evaluation criteria checklists.	MUST
FR-3.7	ROC Review	ROC review scores must be aggregated and presented to the PSR Office for final approval decision.	MUST
FR-3.8	ROC Review	Reviewer assignments must support conflict-of-interest declarations; conflicted reviewers must be automatically excluded from that proposal.	SHOULD
FR-3.9	Monitoring	Approved research projects must have a monitoring module where 6-month progress reports are submitted.	MUST
FR-3.10	Monitoring	The system must track ethical clearance and IRB application status per project.	MUST
FR-3.11	Monitoring	PSR officers must receive alerts for overdue progress reports (past 6-month deadline).	SHOULD
FR-3.12	Final Output	Researchers must be able to submit final policy briefs and full research reports as mandatory outputs.	MUST
FR-3.13	Data Archive	The system must support recording of data deposit references at the NDMC-EPHI central server.	MUST
FR-3.14	Reporting	A research pipeline dashboard (calls → submissions → reviewed → approved → completed) must be available for PSR leadership.	SHOULD
FR-3.15	Discovery	Completed and published research outputs could be made discoverable through the policy repository search.	COULD

3.4 System Administration & Settings

ID	Category	Requirement	Priority
FR-4.1	Taxonomy	Admins must be able to manage organization types, organizations, and units via CRUD interfaces.	MUST
FR-4.2	Taxonomy	Admins must be able to manage thematic areas and strategic objectives used in policy and research tagging.	MUST
FR-4.3	Taxonomy	Policy document types, review statuses, expert decision types, and checklist items must be	MUST

		configurable.	
FR-4.4	Audit Logs	All create, update, delete, and status-change events must be recorded with user ID, timestamp, and before/after snapshot.	MUST
FR-4.5	Audit Logs	Admins must be able to query audit logs by user, date range, module, and action type.	MUST
FR-4.6	Notifications	The system must support configurable notification settings per user (email, in-app, SMS).	SHOULD
FR-4.7	Notifications	System-wide announcements must be postable by administrators.	SHOULD
FR-4.8	Contact	A contact message module must allow external users to submit inquiries to the PSR Office.	COULD
FR-4.9	Dashboard	The admin dashboard must display active users, pending reviews, open calls, and audit activity.	MUST
FR-4.10	Backup	The system should support scheduled data export/backup of all critical records.	SHOULD

4. Non-Functional Requirements

4.1 Security

Ref	Requirement
NFR-S1	All data in transit must be encrypted using TLS 1.2 or higher; HTTP must redirect to HTTPS.
NFR-S2	Passwords must be stored using a strong adaptive hashing algorithm (e.g., bcrypt or Argon2); plaintext passwords must never be stored or logged.
NFR-S3	JWT tokens must have short expiry (access: ≤ 15 min); refresh tokens must be rotated and stored in an HttpOnly cookie where technically feasible.
NFR-S4	The system must implement rate limiting on authentication endpoints (login, OTP generation) to mitigate brute force attacks.
NFR-S5	All API inputs must be validated and sanitized to prevent SQL injection, XSS, and CSRF attacks.
NFR-S6	OTP codes must be cryptographically secure (CSPRNG), 6 digits, with a maximum validity window of 10 minutes and single-use enforcement.
NFR-S7	RBAC must be enforced at the API layer, not solely in the UI; all endpoints must validate permissions server-side.
NFR-S8	File uploads must be validated for type (whitelist), size (max configurable, default 20 MB), and scanned for malware where possible.
NFR-S9	Audit logs must be immutable; no user (including admins) may delete or alter audit log entries.
NFR-S10	Sensitive PII fields (phone, email) must be masked in logs and non-admin API responses unless explicitly needed.

4.2 Performance

Ref	Requirement
NFR-P1	Page load time for dashboard and list screens must not exceed 3 seconds on a standard 4G mobile connection (measured at P95).
NFR-P2	API response times for list endpoints must not exceed 500 ms under normal load (≤ 50 concurrent users).
NFR-P3	File download (policy documents) must initiate within 2 seconds for files up to 50 MB.
NFR-P4	The system must support at least 200 concurrent authenticated users without performance degradation.
NFR-P5	Search queries across the repository must return results within 2 seconds for datasets up to 10,000 documents.
NFR-P6	Notification dispatch must complete within 60 seconds of the triggering event for in-app and email channels.

4.3 Scalability & Availability

Ref	Requirement
NFR-A1	The system architecture must support horizontal scaling of application servers without architectural redesign.
NFR-A2	The platform must target 99.5% monthly uptime (excluding planned maintenance windows announced 48 hours in advance).
NFR-A3	The database must support automated daily backups with point-in-time recovery (minimum 30-day retention).
NFR-A4	The system must handle data growth gracefully — designed to support a minimum of 50,000 documents and 5,000 users within 5 years.
NFR-A5	Planned maintenance must be scheduled during off-peak hours; a maintenance mode page must be shown to users.

4.4 Accessibility & Usability

Ref	Requirement
NFR-U1	The UI must conform to WCAG 2.1 Level AA accessibility standards for color contrast, keyboard navigation, and screen reader support.
NFR-U2	The platform must be fully functional on modern browsers: Chrome, Firefox, Edge, and Safari (latest two versions).
NFR-U3	The UI must be responsive and usable on mobile devices (min screen width: 375px) and desktop (min: 1280px).
NFR-U4	Core workflows (login, submission, review) must be completable without requiring more than 5 clicks from the home screen.
NFR-U5	The system must support both Amharic and English content entry in text fields; UI language should default to English with Amharic option roadmapped.
NFR-U6	Error messages must be human-readable, specific, and provide actionable guidance; generic '500 error' messages must not be exposed.

4.5 Maintainability & Compliance

Ref	Requirement
NFR-M1	The backend must expose OpenAPI/Swagger documentation for all API endpoints to facilitate frontend development and third-party integration.
NFR-M2	Backend and frontend code must include automated test coverage $\geq 70\%$ for critical paths (auth, submission, review, approval).
NFR-M3	The system must comply with Ethiopian national data sovereignty requirements — all data must be hosted on MoH-approved infrastructure within Ethiopia.

NFR-M4	Dependency versions must be tracked; security patches must be applied within 30 days of a critical CVE disclosure.
NFR-M5	System configuration (roles, permissions, taxonomy) must be manageable through the admin UI without code changes.

5. Key User Stories

The following five user stories capture the most critical end-to-end interactions. Each story includes acceptance criteria and links to functional requirements.

US-01 Submit a Research Proposal **MUST**

Persona: Researcher (PI) **Linked FRs:** FR-3.1, FR-3.2, FR-3.3, FR-3.4

User Story

As a researcher, I want to submit a proposal against an open Call for Proposals — including my team details, methodology, and budget — so that my research can be formally considered for MoH funding and support.

Acceptance Criteria

1. Given an open Call for Proposals exists, the researcher can navigate to 'Submit Proposal' from the dashboard.
2. The submission form must capture: PI name, Co-PI details, title, executive summary, thematic area, methodology, and file attachments.
3. The system validates all required fields before allowing submission; field-level error messages are displayed inline.
4. On successful submission, the researcher receives an in-app notification and email confirmation with a reference number.
5. The proposal appears in the PSR Office queue with status 'Pending Screening'.
6. The researcher can view submission status in real time from their dashboard.

US-02 Complete a Structured Research Proposal Review **MUST**

Persona: ROC Technical Reviewer **Linked FRs:** FR-3.6, FR-3.7, FR-3.8

User Story

As a ROC member, I want to access my assigned proposals and complete a structured scoring checklist with comments, so that the PSR Office has a consistent, evidence-based evaluation to inform approval decisions.

Acceptance Criteria

7. The reviewer's dashboard shows only proposals assigned to them, with deadlines and current status.
8. Clicking a proposal opens the full submission package (documents, PI profile, call reference).
9. The review interface presents structured criteria with numeric scoring and free-text comment fields per criterion.
10. The reviewer can save progress and return before final submission.
11. Once submitted, the review score is locked; corrections require PSR Office intervention.
12. The PSR Office can see all reviewer scores aggregated with individual breakdowns.
13. The system flags if a reviewer has a declared conflict of interest and blocks their assignment.

US-03 Approve and Register a Policy Document **MUST**

Persona: PSR Office Official **Linked FRs:** FR-2.3, FR-2.9, FR-2.10, FR-2.11

User Story

As a PSR Office official, I want to review a finalized policy draft and publish it to the central repository

with a unique registration number, so that the approved document is officially traceable, discoverable, and version-controlled.

Acceptance Criteria

14. The official can view all drafts that have passed technical review and are pending final approval.
15. The approval screen shows: the document, all reviewer scores/comments, version history, and thematic area tags.
16. On approval, the system auto-generates a serial registration number (e.g., PSR-POL-2026-0042).
17. The published document record includes: effective date, scheduled review date, document type, owning organization, and version number.
18. The prior version (if any) is automatically marked as 'Outdated' and archived — not deleted.
19. The submitter and associated stakeholders receive a publication notification.

US-04 Onboard a New User and Assign a Role **MUST**

Persona: System Administrator **Linked FRs:** FR-1.5, FR-1.7, FR-1.8, FR-1.10, FR-1.11, FR-4.4

User Story

As a system administrator, I want to create a new user account, assign the appropriate role, and set their organizational affiliation, so that they can access only the modules relevant to their function without manual workarounds.

Acceptance Criteria

20. Admins can create a user via the User Management screen with all required fields pre-validated.
21. Role assignment is done by selecting from a configurable role list; permissions preview is shown before saving.
22. The new user receives a one-time password via email/SMS and is prompted to reset it on first login.
23. The user's account status can be toggled (Active/Inactive) without deleting the account or its history.
24. All account creation, role assignment, and status change events appear in the Audit Log.
25. Admin can search users by name, organization, role, or status with results in under 1 second.

US-05 Monitor the Research and Policy Pipeline **SHOULD**

Persona: Senior Leadership / Director **Linked FRs:** FR-2.13, FR-3.14, FR-4.9

User Story

As a senior MoH director, I want to view a real-time dashboard showing the status of all active policy and research workflows, so that I can identify bottlenecks, track compliance with review timelines, and make informed resource decisions.

Acceptance Criteria

26. The dashboard loads within 3 seconds and displays: total active policies by stage, total proposals by stage, overdue reviews, and recently published documents.
27. Clicking any metric drills into a filtered list view.
28. The dashboard can be filtered by thematic area, organization, and date range.
29. A downloadable executive summary report (PDF/Excel) reflects the current dashboard state.
30. Data shown is refreshed in real time (or within a maximum 5-minute cache window).
31. The director sees only summary data; individual document content requires navigating to the detail view.

6. Open Questions & Clarifications Needed

The following five questions require stakeholder input to finalize this specification. Unresolved items may impact scope, timeline, and technical design decisions.

Q-01 What is the intended approval authority model for policy documents — and can a single role both submit and approve?

Context

Currently, the system separates the submitter and approver roles, but it is unclear whether a PSR Office official could both initiate a policy concept note and approve it. This has direct implications for the RBAC model design and audit trail integrity. Many governance frameworks explicitly forbid self-approval to prevent conflicts of interest.

Decision Impact: *High — affects RBAC design, approval workflow logic, and audit compliance requirements.*

Q-02 Is Amharic language support required for the UI, or only for document content entry?

Context

The platform will serve users primarily in Ethiopia where Amharic is the official working language. It is unclear whether the application interface itself (menus, labels, forms) must be localized to Amharic in v1.0, or whether English-only UI is acceptable with Amharic allowed in text input fields. Full localization is a significant development effort.

Decision Impact: *Medium-High — affects frontend architecture, content strategy, and development timeline by potentially 4-6 weeks.*

Q-03 What are the exact data integration requirements with the NDMC-EPHI central server for research data deposit?

Context

The requirements reference integration with the NDMC-EPHI data server for archiving final research outputs. It is not yet clear whether this means: (a) simply recording a reference number/URL in the PSR platform, (b) a file-upload API integration with NDMC-EPHI, or (c) automated metadata synchronization. Each option has very different technical complexity and timeline implications.

Decision Impact: *High — could require external API contracts, data governance agreements, and significant backend integration work.*

Q-04 What is the policy for handling withdrawn or rejected proposals — should submitters be able to resubmit, and does previous submission history remain visible?

Context

The current specification covers the forward path (submission → review → approval) but does not define behaviour for rejected or withdrawn submissions. Specifically: Can a researcher revise and resubmit a rejected proposal? If so, is it treated as a new submission or a revision? Who can see the history of rejected proposals — the submitter only, the PSR office, or all reviewers? This decision affects data model design and audit trail completeness.

Decision Impact: *Medium — affects workflow state machine design, data retention policy, and user experience for researchers.*

Q-05 What are the deployment infrastructure constraints — is the system expected to run on MoH-managed on-premise servers, a government cloud environment, or a commercial cloud provider?

Context

Several non-functional requirements (data sovereignty, backup, uptime, scalability) have significantly different implementation approaches depending on whether the platform is deployed on-premise, on a government cloud (e.g., EthioTelecom infrastructure), or a commercial cloud with an Ethiopian data residency guarantee. The backend documentation references Django/Gunicorn, suggesting a Linux server deployment, but the target environment is not specified.

Decision Impact: *High — directly determines infrastructure architecture, DevOps toolchain, backup strategy, and cost model for the entire platform.*

Appendix A: Revision History

Version	Date	Author	Summary of Changes
1.0	May 7, 2026	PSR Project Team	Initial draft created based on backend.md and FRONTEND_IMPLEMENTATION_GUIDE.md and project brief.
1.1	TBD	—	Pending stakeholder review and open question resolution.

Appendix B: Requirement Traceability Matrix (Summary)

The following matrix maps each user story to its supporting functional requirements and the primary persona served.

Story	Title	Persona	Functional Requirements
US-01	Submit Research Proposal	Researcher (PI)	FR-3.1 / FR-3.2 / FR-3.3 / FR-3.4
US-02	Complete ROC Review	ROC Reviewer	FR-3.6 / FR-3.7 / FR-3.8
US-03	Approve & Register Policy	PSR Official	FR-2.3 / FR-2.9 / FR-2.10 / FR-2.11
US-04	Onboard User & Assign Role	System Admin	FR-1.5 / FR-1.7 / FR-1.8 / FR-1.10 / FR-1.11 / FR-4.4
US-05	Monitor Pipeline Dashboard	Senior Leadership	FR-2.13 / FR-3.14 / FR-4.9